A Deep Learning-Based Approach for Engagement Assessment of Students with Autism Spectrum Disorder

Md. Aminul Islam Shanto, Sraboni Ghosh Joya, M. Shamim Kaiser, and Md. Sazzadur Rahman

Abstract Engagement assessment plays a crucial role in understanding and improving the educational experiences of students with autism spectrum disorder (ASD). Traditional methods of engagement assessment heavily rely on subjective observations and manual coding, which can be time-consuming, prone to bias, and lack of precision. In this study, we propose a deep learning-based approach, specifically convolutional neural networks (CNNs) to automatically analyze multimodal data streams, including facial expressions and body movements collected during classroom activities. The multimodal data is preprocessed and fed into CNN for feature extraction, followed by CNN-2 for temporal modeling and engagement prediction. The results demonstrate that the proposed deep learning-based approach achieves significantly higher accuracy that is 97.1%. In the future, an Intelligent Tutor System (ITS) will be built to help children with ASD study more effectively via the use of mobile or online applications. Evaluating participation from since engagement is a cognitive trait shared by humans, visual signals alone are insufficient. Physiological elements must thus be present. In the future, we will work on a model that can evaluate a student's degree of involvement by combining behavioral and physiological data.

Keywords Autism spectrum disorder · Students · Classroom · Video data · CNN · Intelligent Tutor System

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1 Introduction

1.1 Overview

A neuro-developmental ailment known as autism spectrum disorder (ASD) is characterized by ongoing difficulties in social interaction and communication as well as the prevalence of limited and repetitive activities. It affects persons at all ages, with symptoms often first emerging in the early years of life. The intellectual and social growth of individuals with ASD is greatly supported by education. However, because of the distinctive cognitive and behavioral traits linked to the illness, it may be especially difficult to include these individuals in educational activities. It helps to gauge students with ASD's extent to participation for a number of reasons. First, the extent of participation, focus, and interest in learning activities may be determined by looking at their level of engagement. It demonstrates their engagement, operation, and capacity for sustained attention, all of which are essential for creating engaging learning environments. Second, measuring engagement may provide important information into how well instructional techniques and treatments designed for students with ASD work. It enables teachers to keep track of student progress, spot areas for development, and modify their pedagogical approaches as necessary. Finally, engagement assessment contributes to inclusive education by ensuring that children with ASD get the proper assistance and modifications based on their requirements.

1.2 Limitations of Existing Works

Traditional methods of engagement assessment in students with ASD primarily rely on subjective observations, self-reporting, and behavioral checklists. These methods are often labor-intensive, time-consuming, and prone to inter-rater variability. Moreover, they may not capture the subtle nuances and context-specific cues associated with engagement in individuals with ASD. Consequently, there is a need for objective, reliable, and efficient assessment methods that can provide accurate and real-time insights into the engagement levels of students with ASD.

1.3 Motivation of This Study

Deep learning's potential to autonomously recognize and separate significant patterns and characteristics from large amounts of complicated data precisely propels its use in engagement assessment. Large volumes of data, such as the heterogeneous sensor data and video recordings that are frequently employed in engagement assessment, are particularly adequate for processing by deep learning algorithms. By using deep

learning, we can create models that can identify complex patterns and nuanced indicators related to participation in kids with ASD, resulting in more precise and impartial evaluations. Deep learning models also have the aptitude to make assumptions and respond effectively to various situations and individual characteristics, making them especially appropriate for assessing engagement in a diverse population like students with ASD.

1.4 Our Contribution

The research involved was motivated by the dearth of educational resources for students with ASD. In addition, we need to make sure that students with ASD do not slack off and fall behind in their studies for no apparent reason. In this study, we provide a deep learning-based method for assessing engagement among students with autism spectrum disorder. Our goal is to provide teachers and researchers with a robust and practical method to assess interest now by capitalizing on the power of deep learning algorithms. This strategy has the potential to help individuals with ASD succeed in school by facilitating individualized treatments and establishing welcoming classroom settings. Using the camera to capture the children's reactions while the teacher was talking allowed us to generate a dataset that was then interpreted by a specialist in autism spectrum disorders. By using deep learning algorithms, we have differentiated between four stages of student engagement. Highly involved, partially engaged, not engaged, and undetermined are the four categories of participation.

2 Related Work

With the advancement of artificial intelligence (AI) through deep neural networks, automated engagement analysis has gained traction since the early 2000s. It was made possible by the emergence of big data, which is now relatively easy to acquire. If there is another factor that must be addressed, it is undoubtedly the rise in computational power. A classroom-based automated engagement monitoring system was first proposed by Bidwell and Fuchs [1]. After that several methods already in use have been examined by some other researchers. For the purpose of assessing engagement among students with ASD, this section presents an overview of some of these methods, pointing out their benefits, drawbacks, and remaining.

2.1 Overview of Different Methods

Observational Methods: Observational approaches require closely observing how students behave and interact while participating in instructional events. To classify engagement based on behavioral indicators including attention, active involvement, and completion of tasks, teachers and researchers employ conventional classification systems. According to McKenzie et al., there are certain observational approaches that were used to collect rich subjective data for gauging student participation during class, but they are time-consuming, subject to bias from observers, and need extensive training [2]. A classroom observation procedure was created by Lane and Harris to quantify student involvement in big university classrooms [3]. Educators may get real-time data on how their students are reacting to class material using the Behavioral Engagement Related to Instruction (BERI) protocol [3]. After that in 2015 Currey et al. intended to assess the attitudes toward and participation in team-based learning (TBL) among postgraduate nursing critical care students [4]. A comprehensive evaluation of works on student involvement analysis was conducted by Thomas and Jayagopi [5]. A prediction approach that can assess whether students were paying attention or preoccupied based on captured footage from the classroom was also presented [5]. These studies show that observational approaches include direct visual monitoring of students' relationships and behavior while participating in instructional activities.

Questionnaires and Rating Scales: Techniques for personal or caregiver-report which assess several facets of engagement, including interest, focus, and motivation, include questionnaires and rating scales. These methods may record the opinions of people with ASD and provide arbitrary insights regarding participation levels. To evaluate the usage of questionnaires and rating scales in assessing student participation, Fredricks and McColskey suggested a model [6]. Quantitative studies of classroom climate and student participation sit beside qualitative assessment systems for general student conduct [6]. Applying the Rasch–Andrich rating scale model, You et al. provided a method for examining students' levels of enthusiasm and validated the Science Motivation Questionnaire II (SMQ II) created by Glynn et al. (2011) [7]. Many qualitative studies on parental engagement have flaws, and Bakker and Denessen focused on them. On an evaluation of twelve research chosen for their use of quantitative measures of parental participation [8]. The results of this research suggest, however, that self-report assessments might be especially problematic for people who have trouble expressing themselves or who have a shallow understanding of their own capabilities. There is a risk of bias in caregiver reporting. In addition, the responses to surveys and rating scales are generally based on memory, which may not be an accurate representation of participants' involvement in the moment.

Audio and Video Analysis: Analyzing audio and video data is a method used to discover engagement-related activities and signs including talking, smiling, and touching. Computer vision and audio processing methods have been combined with machine learning algorithms to automatically extract features and categorize degrees of involvement. A fresh paradigm for gauging student interest in their classroom or online coursework was presented by Alkabbany et al. [9]. The user's video was recorded, and the suggested framework followed the user's face as it moved from frame to frame [9]. As shown by the student's gestures, glance patterns, and overall achievement during brief video learning sessions, Yue et al. suggested a combined structure to assess engagement with learning from three facets: impact, conduct, and mental state [10]. A technique for gauging pupils' degrees of focus was carried out by Li et al., who used a photoplethysmogram (PPG) signal detected by a device that is worn with a median accuracy of 0.69 [11]. Convolutional neural networks were used in edge computing by Umarale et al. to offer a deep learning method for determining the learners' degree of attention [12]. Students' levels of engagement are divided into one of four categories by the neural network: extremely attentive, attentive, less attentive, and least attentive [12]. An open-source dataset was used in the training of a deep learning convolutional neural network (CNN) created by Wang et al. [13]. Quantitative experiments illustrated the effectiveness of this model and the deep learning CNN in continuous tracking of student progress and engagement. These methods allow for instantaneous, objective evaluation. However, obstacles include a need for reliable algorithms, susceptibility to external influences, and the necessity of sufficient training data.

Physiological Measures: Heart rate variability, electrodermal activity, and eye tracking are just few of the physiological measurements that have been investigated for use in assessing students' levels of involvement who have autism spectrum disorder. Arousal, focus, and mood may all be quantified using these tests. Physiological metrics provide a continual, non-invasive evaluation of effort. They are complicated to interpret because of individual variances and co-occurring disorders, need specialist equipment, and are susceptible to outside influences. GSR detectors were employed as a measure of students' participation in a distributed learning environment, as shown by the work of Wang and Cesar [14]. A self-report questionnaire was developed to evaluate two kinds of student involvement with school: cognitive and psychological engagement, based on the literature review conducted by Appleton et al. [15]. Based on answers from a socioeconomically and racially diverse group of urban ninth graders in 1931, the student engagement instrument's (SEI) psychological characteristics were evaluated [15]. The building of a tool that allows students to self-report to evaluate the two kinds of student involvement with school—cognitive and psychological engagement—was made possible by Fredricks et al.'s study of pertinent literature [6]. Responses from an urban group of 1931 ninth-grade students from a variety of racial and socioeconomic backgrounds were used to evaluate the statistical features of this measure, the SEI [6].

Sensor-Based Approaches: Wearable sensors that capture information about a wearer's position, movement, and interactions with others include accelerometers and proximity sensors. These sensors enable undetectable, impartial evaluations of involvement. The gathered sensor data may be utilized to classify the levels of user interaction using machine learning algorithms. There are limitations, however, such as the need for good sensor implantation, the requirement to guarantee participant comfort, and the requirement to handle data privacy problems. By using the Emotiv EPOC system, Ghergulescu and Muntean suggested a unique EEG sensor-based way to gauge and analyze learners' motivation in game-based learning environments [16]. The study discussed in this article has the potential to promote and allow motivation-based adaptation in educational games by offering an automated real-time solution for motivation evaluation [16]. As trainees advanced through a robotic surgeon training program in a medical facility, Wu et al. suggested a model to track changes in their cognitive and behavioral states [17].

2.2 Limitations and Gaps in Current Literature

It is important to recognize the constraints inherent in each study before drawing definitive conclusions about its reliability or validity. This is more than simply a synopsis of the frequency and trend of inaccuracies and other validity concerns. There has been a lot of focus on engagement analysis recently. Constraints unique to each project must be identified and addressed in the future. The limitations of the studies we looked at are shown in Table 1.

3 Methodology

The deep learning-based model we presented in this part can be seen in Fig. 1. The entire approach has broken down into many phases. The first phase consisted of capturing some unedited films of a school environment. To make it easier for the system to handle those lengthier flicks more rapidly, we then have sliced them into shorter flicks. Then, components of the head and body positions have extracted using an application called Mediapipe. Then, we have then incorporated the preprocessed features to the training of a deep learning model. After a deep learning model has been trained, it is feasible to predict how interested an ASD student will be. There are four different groups in this categorization since it has many classes. The four levels of engagement are highly engaged, moderately engaged, not engaged, and undefined.

Table 1 Limitations in current literature

Authors and year of publication	Title	Limitations
Bidwell and Fuchs 2011 [1]	Classroom analytics: measuring student engagement with automated gaze tracking	They must actively click a start-record button to capture video data; the main limitation is that there are no scheduling features to initiate an automatic video recording
Alkabbany et al. 2019 [9]	Measuring student engagement level using facial information	They used manual annotation, which is often unreliable. It is applied to normal students
Yue et al. 2019 [10]	Recognizing multidimensional engagement of e-learners based on multi-channel data in e-learning environment	Their proposed model needs a huge training dataset
Li et al. 2020 [11]	A learning attention monitoring system via photoplethysmogram using wearable wrist devices	They proposed a definition of attention that is limited by the data available, and the amount of data they labeled as engaged is insignificant
Umarale et al. 2020 [12]	Attention detection of participants during digital learning sessions using edge computing	The method they describe here is not an automated prediction of distance learning engagement. They did not analyze the performance during stressful events experienced by learners, such as quizzes or exams
Wang et al. 2020 [13]	Automated student engagement monitoring and evaluation during learning in the wild	They have used ground truth-based annotation which is not reliable. Using sensors is often expensive, and another aspect is they address the normal student in a classroom setting
Appleton et al. 2006 [15]	Measuring cognitive and psychological engagement: validation of the student engagement instrument	Shortage of data to train models, so the accuracy is not up to mark and they do not have much variation in disengaged samples, so it tends to predict 'engaged' more in educational environment. It does not address students with ASD

3.1 Data Collection and Preprocessing

It is important to recognize the constraints inherent in each study before drawing definitive. The development and implementation of a deep learning-based engagement assessment method for students with ASD requires thorough data collection and preparation. These steps involve gathering relevant data and preparing it for further research and modeling. Below is an overview of the steps involved in collecting and preparing data, and Fig. 2 describes the whole data collection procedure in a summary.

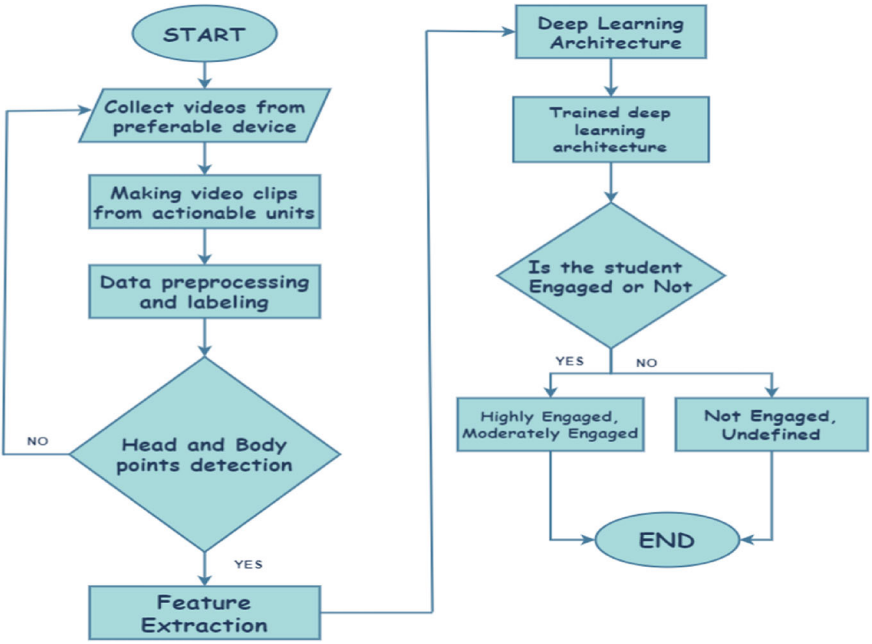


Fig. 1 Flowchart of proposed model

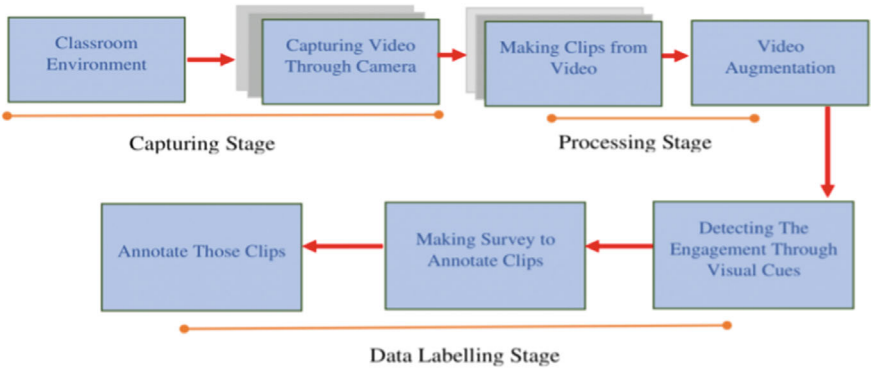


Fig. 2 Procedure of collecting raw video data for the model

a. Capturing videos from the classroom: To gauge how much of engagement of students with ASD, two participants were recorded for a total of 4 h, 34 min while they were receiving instructions from a teacher. Many of the students were already familiar on the key ideas being discussed in class. We were recording on video at the precise moment. Back then, we relied on our mobile cameras. They were cautioned against becoming stale in front of massive video cameras.

b. Video capturing and making short clips: In class, they learned by repeating a few concepts they were already acquainted with. That's when we first began using the camera to shoot video. We took these pictures using the cameras on our mobile phones. Because teachers emphasized the need of avoiding developing an irrational fear of large video cameras. The cameras were pointed slightly to the learner's left rather than directly at them. The recording procedure was carried out in this way without arousing any suspicion.
c. Video preprocessing and augmentation: To use the body and head movements of students as indicators of their mental focus, we need to remove the pertinent segments from the recordings. After that, the model is over fit using augmentation techniques that are utilized to clip regeneration. After that, it is necessary for us to personally examine the standard of the films. As a consequence of this, we now have a total of 3332 clips, of which 825 have been evaluated as 'highly engaged' and 824 have been rated as 'moderately engaged.' There are 829 of them that are categorized as 'not engaged,' and there are 854 more that are 'undefined.'
d. Video annotation: Following the creation of the videos, we are conducting a poll of educators at a school for students with autism to determine the best annotation strategies. They have some suggestions on how to annotate the dataset to identify four labels: 'highly engaged,' 'moderately engaged,' 'not engaged,' and 'undefined' based on the visual cues and our questionnaire.

3.2 Feature Extraction and Representation

To create a deep learning-based solution for engagement evaluation in kids with ASD, feature extraction and representation are essential stages. These procedures include converting the raw data into relevant and meaningful characteristics that may be fed into a deep learning model. The feature extraction and representation procedure are summarized as follows:

1. Detecting the subject: After getting the analyzed motion pictures, we proceeded with these operations. Using the machine vision framework Mediapipe [18], we have performed subject detection. We weren't able to examine every person in the room using the framework since it could recognize many faces at once. After identifying the individual, we have employed 3D models to extract traits like body and head points. The distance between the person being photographed and the camera is indicated by 'Z'. Since a person does not always stand in front of the camera at the same distance, depth measurement is necessary to identify the subject's posture. As a result, Mediapipe is the best detection framework and superior to other frameworks. The model for predicting postures is called BlazePose, and it is being utilized here.
2. Detecting the body points: Following the identification of features, we have continued to pursue the issue in reference to those features. It would locate and keep track of 33 different anatomical landmarks. We were only utilizing upper body landmarks as our key body point to extract features since the nature of our



Fig. 3 Extracted features for building our assessment model

video and the engagement assessment both favored using landmarks that show the position of the head and body. This is because landmarks that indicate the location of the head and body are more likely to be accurate.

3. Extracting the body points: We have removed 11 points from the head point, including the nose, left inner and outer eyes, right eye, and outer eye, left and right ears, and the left and right mouths. We have extracted an additional 12 points from the body point, including the left and right mouths, the left and right shoulders, the left and right elbows, the left and right wrists, the left and right pinkies, the left and right indexes, and the left and right thumbs. Figure 3 indicates the extracted features.
4. Extracting features by using 1-D CNN: 1-D CNN performs better at extracting features from time-series data. A one-dimensional CNN was utilized to extract the feature from the body points in this model. Because video is a collection of frames, it undoubtedly contains a time-series component, and by applying the convolution layer, it can detect and retrieve the feature that binds the inputs to the outputs directly.

3.3 Deep Learning Architecture for Engagement Assessment

In the present study, CNN-2 fared better than the other models; hence, we intend on using the CNN-2 architecture as the basis for our deep learning system. Our system architecture is visualized here in Fig. 4. Raw video is recorded in this model using a resolution of 1920 by 1080 pixels and a frame rate of 24 frames per second. After the videos have been captured, shorter clips are created with the goal to reduce the amount of computing effort required for the process. After that, the clips have been cleaned up and labeled to ensure that the quality is maintained throughout. Following the annotation, a pre-trained architecture known as Mediapipe is used to extract essential information from both the head and the body of the document. It is put to

use in the process of separating the important points from the body points and the head points. After that, information is input into the convolution layer, which consists of 64 neurons and uses the 'Tanh' activation function. After this stage, it proceeds to the next one, which is a max-pooling layer that halves the input. Following that, a dropout layer consisting of 25% is added so that there is no overfitting. Following this dropout layer is a new convolution layer that has 128 neurons and uses the 'Tanh' activation function as its activation function. This is followed by a max-pooling layer and a dropout layer. A layer containing 256 neurons and an activation function denoted as 'Tanh' (hyperbolic tangent function) is added as the third convolution layer of the network. After going through the max-pooling and dropout layers, the data is then flattened before being fed into fully linked dense layers. This function, known as Tanh, is responsible for activating these layers. After that, at the output layer, the softmax activation function is employed to make a prediction about the kind of engagement. As a loss function, categorical cross-entropy was used in the construction of this model.

3.4 Model Train

The development of a deep learning-based strategy for engagement assessment in students with ASD must include a model training stage. We have applied four models. They are artificial neural network (ANN), convolution neural network-1 (CNN-1), long short-term memory (LSTM), convolutional neural network-2 (CNN-2, with different architecture).

ANN: It consists of 13 layers, where 5 of them are dense layers with 64, 128, 256, 128, and 64 neurons in each layer, followed by a dropout layer with a 25% chance of omitting the neurons to activate the next layer. Dropout layers are used to prevent the overfitting of the model. In between the third and fourth dense layers, batch normalization is introduced so that the values of weighted sum operation do not explode, and all numbers belong to the same range. The activation function rectified linear unit (ReLU) has been used. Categorical cross-entropy has been used as a loss function as this is a multi-class classification problem and adaptive moment estimation (Adam) is the optimizer. For the same reason, 'softmax' is used as an activation function in the output layer. This layer has four outputs—'highly engaged,' 'moderate engaged,' 'not engaged,' and 'undefined.'

CNN-1: There are two convolution layers, followed by three fully connected layers with the activation function being 'ReLU.' In the first convolution layer, it has 128 neurons, and in the second, there are 256 neurons with kernel size 3 for both layers. There is a dropout layer in both convolution layers, followed by a max-pooling layer, which reduces the input size and extracts the essential features. After the flatten layer, there are three dense layers with 256, 128, and 64 neurons. Softmax activation function is used in the output layer for multi-class classification, and categorical cross-entropy is used as a loss function while Adam is the optimizer.

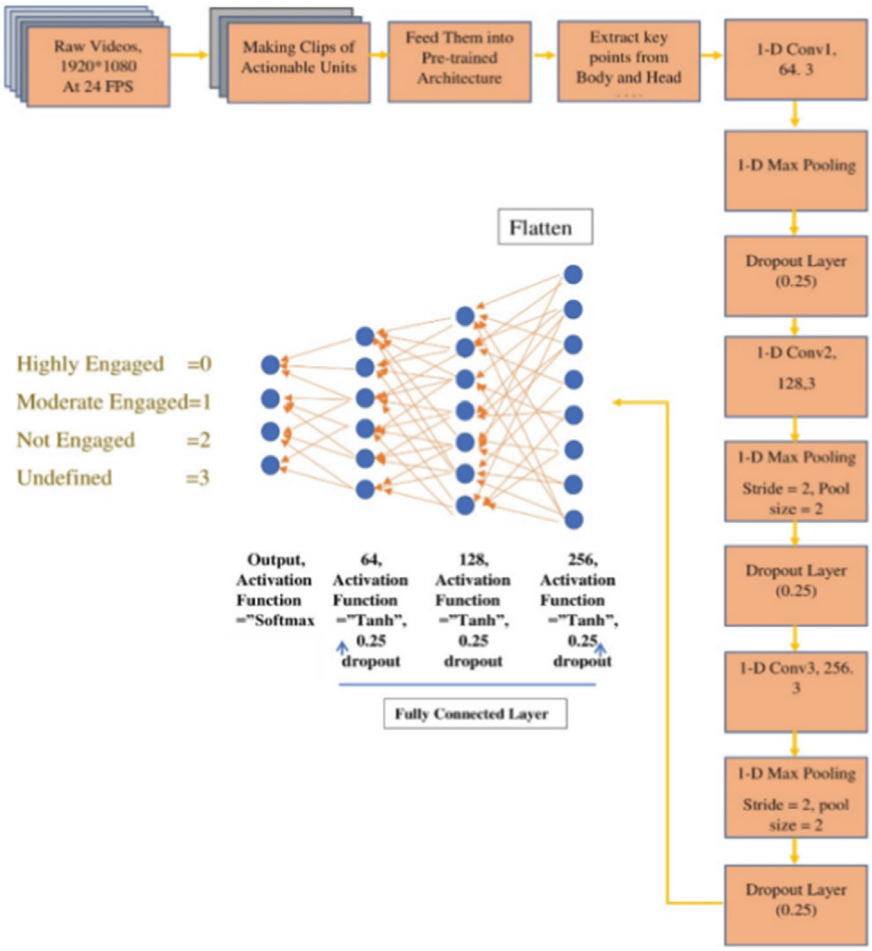


Fig. 4 System architecture of our assessment model

LSTM: Since LSTM works better for time-series analysis, this architecture is proposed for this study. There are 4 LSTM layers with 64, 128, 256, and 512 neurons, respectively, with the activation function being 'Tanh.' After the fourth LSTM layer, a dropout layer is introduced to prevent the model overfitting. There are four dense layers with 512, 256, 128, and 64 neurons, respectively, with the activation function being rectified linear unit (ReLU) followed by dropout layers with 25% probability. Categorical cross-entropy is being used as a loss function. In the output layer, softmax is used as an activation function. Adam is used as an optimizer for this model.

CNN-2: This architecture comprises a total of 16 layers. There are three convolution layers with 64, 128, and 256 neurons with a kernel size of 3. 'Tanh' is used as an activation function for these convolution layers. Each convolution layer has a

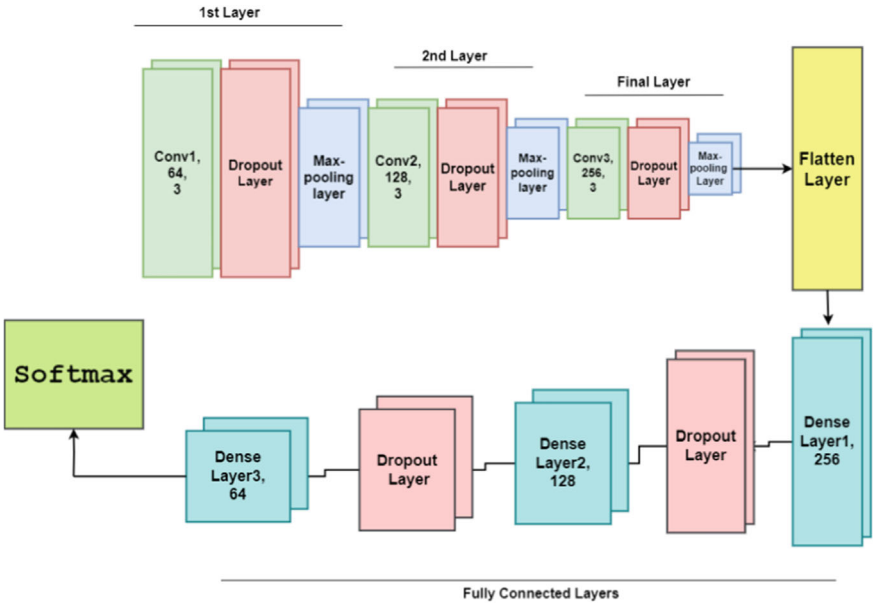


Fig. 5 CNN-2 model that outperforms better than other selected models

dropout layer with a 25% possibility and a max-pooling layer to make the operation less computationally intensive and get the essential features. There are three fully connected layers with 256, 128, and 64 neurons. 'Tanh' is used as an activation function. After the first and second dense layers, dropout layers with a 25% possibility are introduced to prevent the model from overfitting. Softmax is used as an activation function for the output layer, while RMSprop is used as an optimizer. Since it is a multi-class classification problem, categorical cross-entropy is used as a loss function. The whole CNN-2 model is illustrated in Fig. 5.

4 Dataset

The dataset underlying the study's engagement evaluation of students with ASD is described in this section. It contains information on the procedures used to acquire the data, participant demographics, and the inter-rater reliability of the annotation process.

Description of the Dataset: Since the dataset has split into a 75:25 train-test ratio, it has a total of 833 clips for test and 2499 clips to train the model. Since the dataset has 4 classes, it is a multi-class classification problem. The number of clips for train and test dataset split is in Fig. 6.

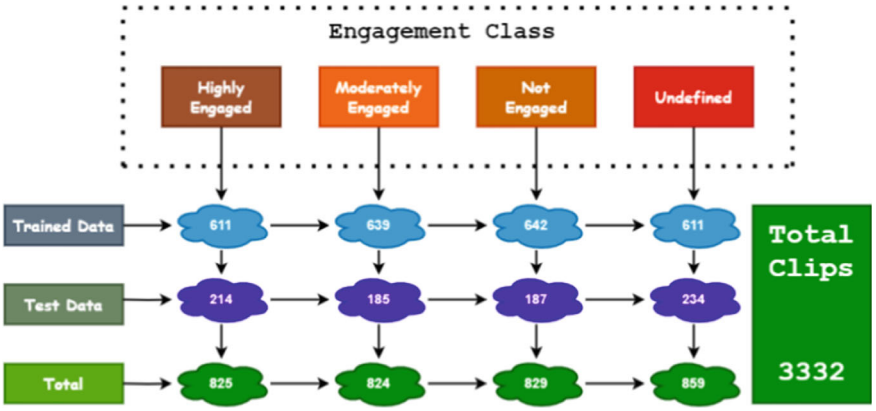


Fig. 6 Annotated dataset that has been used for our model

Participant Demographics: Working with educational institutions, special education programs, and pertinent stakeholders was a part of the data gathering procedure. Students with an ASD diagnosis who were between the ages of 6 and 12 made up the study's subjects. Before any data was collected, the participants' parents or legal guardians gave their approval. To guarantee ecological validity, the data collecting sessions were conducted in environments designed for learning, such as schools or therapeutic spaces. Several recording tools, such as video cameras, wearable physiological sensors, and audio recorders, were employed to record several modalities concurrently. The participants took part in a variety of educational assignments that were created to elicit varying degrees of involvement, such as group discussions, one-on-one conversations, and instructional exercises. The data gathering sessions were rigorously watched by researchers and educators to guarantee participant comfort and reduce any negative impacts. When necessary, breaks and downtime were given, and stringent privacy and confidentiality rules were followed.

Annotation Method and Inter-Rater Reliability: Professional annotators with in-depth knowledge of ASD research and engagement evaluation annotated the gathered data. The data was divided into smaller segments throughout the annotation process, such as time frames or activity-based segments, and each segment was given an engagement label. Multiple annotators individually tagged a portion of the data to assure consistency and dependability in the annotation process. To evaluate the agreement between the annotators, inter-rater reliability metrics such as Cohen's kappa coefficient and intraclass correlation coefficient (ICC) were calculated. To rectify any differences or ambiguities in labeling requirements, the annotators had frequent meetings and discussions. To reduce subjectivity and improve consistency, the annotation rules gave precise definitions and criteria for each type of engagement level (highly engaged, moderately engaged, not engaged, and undefined). Through consensus-based talks or the inclusion of a third expert to make the ultimate decision, any discrepancies or doubts were addressed. By contrasting the annotations made by

the various annotators, the inter-rater reliability was evaluated. The engagement labels given to the dataset were consistent and of excellent quality, according to the reliability assessments that followed, which showed a high degree of agreement.

5 Result and Discussion

This section uses a variety of evaluation matrices and graphs to show and evaluate the experimental data for the provided models. Here, we show the model accuracy as well as the validation loss for each model that we implemented and calculated the validation accuracy for each model. Here, we also provide the confusion matrix considering CNN-2 architecture, as well as precision, recall, and F1-score for a specific class in relation to the suggested models.

5.1 Model Accuracy and Validation Loss

We started by figuring out the validation accuracy. Four alternative suggested architectures have achieved validation accuracy after 100 complete epochs. Next, we established the model accuracy that was previously employed in this research. Figures 7 and 8 indicate the validation accuracy after 100 epochs as well as the accuracy of used models and their area under the roc curve (AUC).

ANN: After fine-tuning the hyperparameters on a trial-and-error basis, it achieved an accuracy of 94.4% and AUC of 0.988 and with 5 fully connected dense layers, consisting of 64, 128, 256, 128, and 32 neurons, with an activation function of

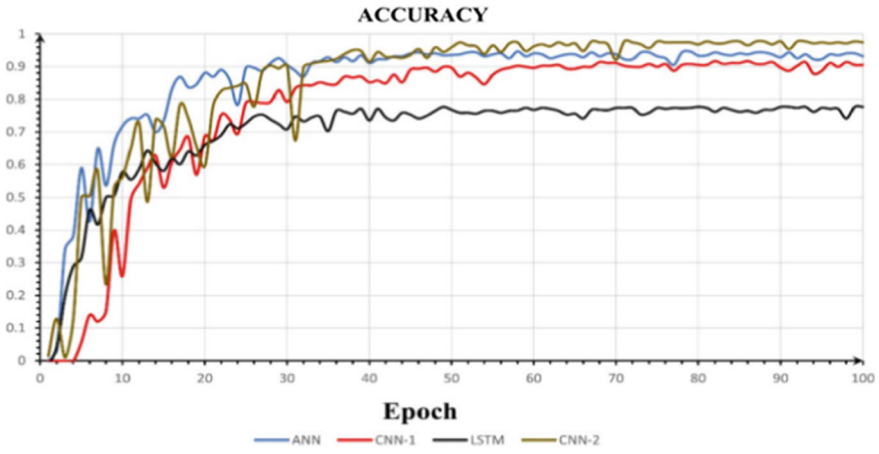


Fig. 7 Validation accuracy after 100 epochs

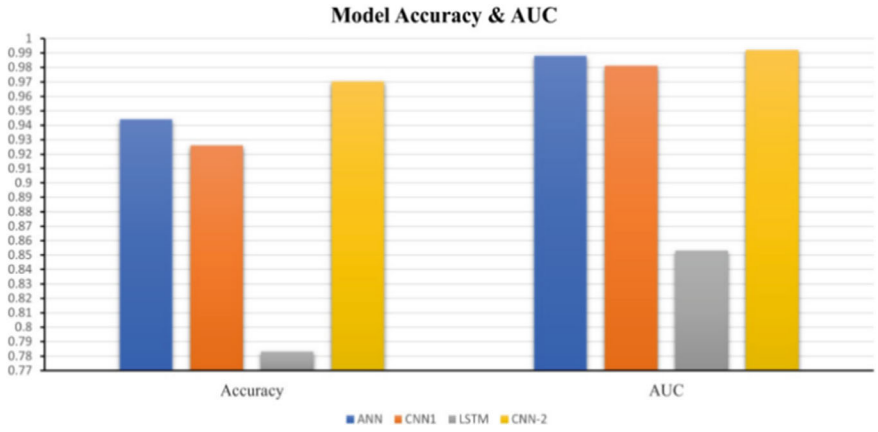


Fig. 8 Model accuracy and AUC

rectified linear unit (ReLU). It has categorical cross-entropy as loss function and adaptive moment estimation (Adam) is the optimizer. The output layer has a softmax activation function percent function.

CNN-1: After fine-tuning the hyperparameters on a trial-and-error basis, it achieved the accuracy of 94.4% and AUC of 0.988, with 3 fully connected dense layers, and 2 convolution layers, consisting of 128 and 256 with an activation function of rectified linear unit (ReLU). It has categorical cross-entropy as loss function and adaptive moment estimation (Adam) is the optimizer. The output layer has a softmax activation function.

LSTM: After fine-tuning the hyperparameters on a trial-and-error basis, it achieved the accuracy of 78.3% and AUC of 0.852, and, with 3 fully connected dense layers, and 3 LSTM layers, consisting of 64, 128, and 256 with an activation function of rectified linear unit (ReLU). It has categorical cross-entropy as loss function and adaptive moment estimation (Adam) is the optimizer. The output layer has a softmax activation function.

CNN-2: After fine-tuning the hyperparameters on a trial-and-error basis, it achieved the accuracy of 97.1% and AUC of 0.992, and, with 3 fully connected dense layers, and 3 convolution layers, consisting of 64, 128, and 256 with an activation function of hyperbolic tangent function (Tanh). It has categorical cross-entropy as loss function and root-mean-squared-propagation is the optimizer. The output layer has a softmax activation function.

From this assessment, it can be stated that CNN-2 has outperformed other models. Since CNN-2 has 3 convolution layers and 3 fully connected dense layers, it has better accuracy than the rest. CNN-1-D typically works better in time series, and it has been proved here too. Though the second model was CNN-1-D too, it has suffered from fewer convolution layers and fully dense layers. LSTM is another one that is better

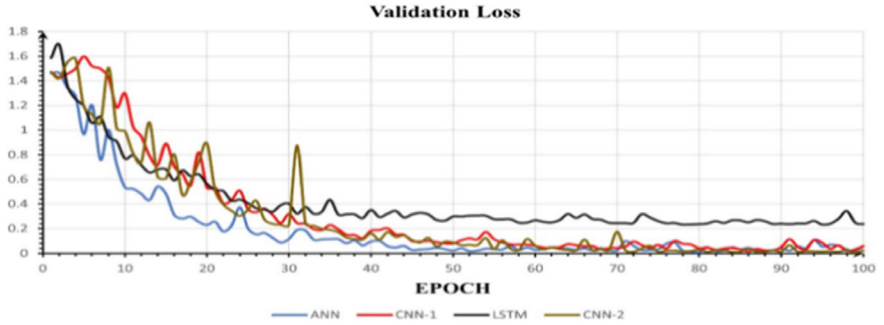


Fig. 9 Validation loss for different models

for time series but seems like it has suffered computationally. We may need more layers of LSTM. ANN typically works well for these kinds of problems, and we certainly see this here too. It has outperformed CNN-1 architecture but falls behind CNN-2.

Validation Loss: According to Fig. 9, in ANN, CNN-1, LSTM, it looks like the validation loss converges after 40–50 epochs. It would be good if we stopped the test at that time. These three have used adaptive moment estimation (Adam) as an optimizer. So, it is understandable why the loss function doesn't fluctuate more and converge quickly.

If we observe the validation loss of CNN-2 which uses root-mean-squared-propagation as an optimizer, validation loss can certainly imply that it converged later than adaptive moment estimation (Adam) but it gives less validation error/loss over time than the other three.

5.2 Precision, Recall, F1-Score for Particular Class in Respect to the Proposed Models

From Fig. 10, we can describe that in the 'highly engaged' class, CNN-2 outperforms other neural network architecture. It has scored 0.98 in precision, which means it can easily project the output correctly in 98% time. In recall, 97% of the time, CNN-2 can get the correct class from the predicted class. In terms of the F1-score, it works best from the rest of the models. So, it can be easily certified that CNN-2 is better neural network architecture among others in predicting the 'highly engaged' class.

In the 'moderate engaged' class, CNN-2 outperformed others with a score of 0.97 in precision which means only predicts wrong only 3 times in a hundred times. In recall, it can predict the true class 99% of time, while the F1-score is 96%. LSTM stands last in every criterion whereas CNN-1 and ANN belong second and third in every matrix. It can certainly be implied that CNN-2 has predicted a better outcome than the rest of the neural network architectures in the 'moderate engaged' class.

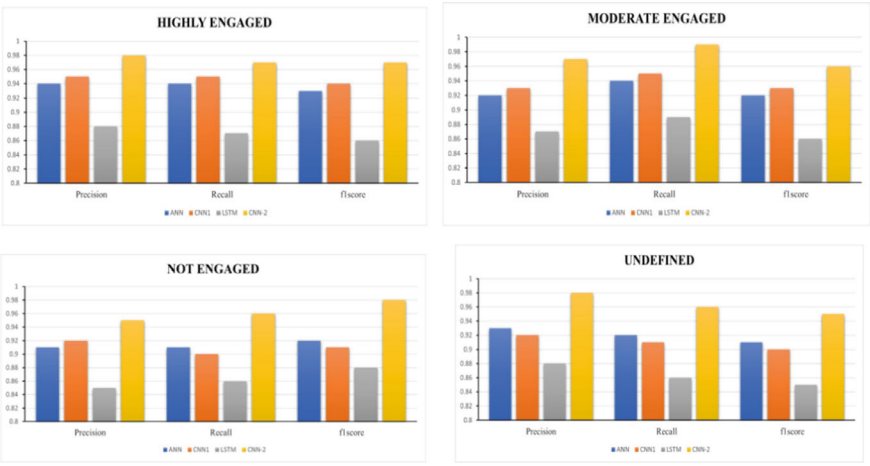


Fig. 10 Precision, recall, F1-score for highly engaged, moderately engaged, not engaged, and undefined

In the 'not engaged' class, whether the student is engaged or not, CNN-2 outperformed others with a score of 0.95 in precision, which means it can predict the student is not engaged 95 times out of 100 samples. Since the recall score is 0.96, it can predict the actual class 96% of the time and 0.98 in the F1-score. In this regard, ANN stands second in recall and F1-score, while CNN-1 outperforms ANN in the precision matrix.

In the 'undefined' class, CNN-2 outperformed others with a score of 0.98 in precision, which means it can predict the true class 98 times out of 100 times from the predicted true class. It can predict the 'undefined' class as a true class 96 times out of 100 times from the predicted class, and 0.95 in the F1-score. In this 'undefined' class, ANN stands second in every matrix while LSTM stands last.

5.3 Confusion Matrix

A confusion matrix gives a brief comparison of the model's predictions and the actual results. The matrix comprises four cells: true positives (instances of positive outcomes predicted correctly), false positives (instances of positive outcomes predicted wrongly), true negatives (instances of negative outcomes predicted correctly), and false negatives (instances of negative outcomes predicted incorrectly). The confusion matrix that we obtained from our model is shown in Fig. 11. It aids in evaluating a classification model's accuracy, precision, recall, and other performance indicators. Since the dataset splits into 75:25 for train-test. In the test folder, there are 833 clips to evaluate the model. Out of 833 clips, 211 clips are 'Highly Engaged' class, 188 clips belong to the 'Moderate Engaged' class, 193 clips are labeled as

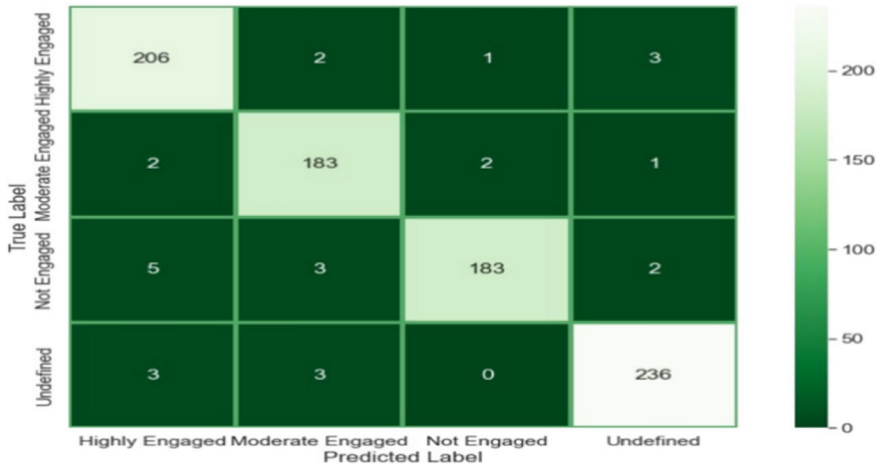


Fig. 11 Confusion matrix for our system model

'Not Engaged,' and the rest of the 241 clips are 'Undefined.' Out of 211 clips in the 'Highly Engaged' class, 206 clips are predicted correctly by the model. 183 of 188 clips from the 'Moderate Engaged' class are predicted correctly, 183 of 193 clips from the 'Not Engaged' are defined correctly by the model, and 237 of 241 clips are correctly predicted by the model as 'Undefined.'

5.4 Discussion

After observing results from validation accuracy, loss, AUC, and class-wise performance, it is certainly implied that CNN-2 has performed better than the rest of the models. It has 3 convolution layers that can essentially extract the features and then the fully connected layers can identify patterns between the input and output. To find specific patterns, ANN has a reputation for doing a good job. In this case, it can certainly be stated that it has done a better job by placing itself in the second position out of four proposed models. LSTM stands last in every category while it generally works better in time-series components. The reason could be that since it is a very computationally intensive model, it needs more data to learn. CNN-1, which has two layers of convolution layers to extract features, can certainly do better if it has more convolution layers. Since, in this study, four models are proposed, we want to see how the models perform if the layers and the parameters are reduced or increased. CNN-1 could perform better if there was an extra convolution layer. Train accuracy is omitted for every model because, in the four proposed models, there are different parameters used so the train accuracy cannot be comparable among these four models.

6 Limitations of This Study

The limited sample size used to train and evaluate the deep learning model is a limitation of this study. Results would be more reliable and generalizable with an increased sample size. The results may not be applicable to other situations or populations since the research may have been carried out in a particular environment or with a particular group of students. Additionally, the research could not have taken into consideration possible confounding factors such individual variations in sensory processing or coexisting illnesses that might affect student participation. To confirm and expand on the results, further research is required using bigger and more differed categories.

7 Conclusion

Consequently, there is considerable potential for improving educational treatments and support for people with ASD. A deep learning-based approach to engagement evaluation in students with ASD has demonstrated encouraging outcomes. Advanced deep learning algorithms combined with multimodal data integration have made it possible to monitor student involvement in a way that is more accurate and objective, which has improved our comprehension of their learning styles and demands. Scalability is a benefit of the deep learning-based technique since it may be used in a variety of educational contexts and can handle enormous datasets. This creates opportunities for the long-term observation of student participation and the gathering of copious amounts of data that may aid in continuing research and the development of educational strategies for students with ASD. To divide the students' degree of participation into four groups, we created a study project using CNN-based architecture. There are four degrees of engagement: highly engaged, moderately engaged, not engaged, and undefinable. In this study, we built an annotated dataset using behavioral cues. We created a dataset using real classroom footage, tagged it, and trained it using clips that ranged in length from 3 to 8 s. The videos that were acquired by the camera are used in this way to extract data about the body and head attitude. In our suggested method, a deep learning CNN classifier is built using student video from the dataset and then utilized for monitoring and evaluating the participation of ASD students. The experimental findings show that the model has a 97.1% accuracy rate. According to test findings, this approach for determining students' levels of involvement based on visible student behaviors in the classroom setting yields a respectable identification rate and favorable outcomes.

8 Future Work

In this research, we primarily concentrate on the engagement categorization for kids with ASD prediction side. A collection of 3332 annotated video clips has been created. But it is useless in a classroom setting. Based on this approach, an Intelligent Tutor System (ITS) will be built in the future to help children with ASD study more effectively via the use of mobile or online applications. Evaluating participation from since engagement is a cognitive trait shared by humans, visual signals alone are insufficient. Physiological elements must thus be present. In the future, we will work on a model that can evaluate a student's degree of involvement by combining behavioral and physiological data.

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